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RESEARCH ARTICLE

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Exploring the implications of cloud-based enterprise resource planning systems for public sector management accountants

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Abstract

Based on a case study of a large Swedish local government municipality, we explored the extent to which a cloud-based enterprise resource planning (CERP) system enabled the role performance of public sector management accountants. Our findings suggest that the CERP system enabled central management accountants to mobilize their specific expertise because it eliminated manual work, increased transparency, and made them feel more comfortable with the numbers. However, the less flexible features of a CERP system provided by external vendors, such as limited customization, posed a challenge for the local management accountants serving the different needs of a diverse range of managers and business units. Looking at the different attitudes that central and local management accountants developed toward the CERP system, we found that although both focal groups in our analysis belonged to the same occupation, they framed the role of technology differently. Although local management accountants framed it as a tool that should enable them to draw on their local expertise to produce tailor-made information for their local units, central management accountants saw the CERP system as a tool that allowed them to consume prefabricated "high-quality" information to assure efficient and risk-free accounting

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processes throughout the entire organization. Given this, cloud technology constitutes a risk that accounting and control processes become unduly inflexible and cumbersome at the local level. Coping with an inflexible cloud-based system may therefore add to the list of challenges that public sector management accountants experience when trying to be(come) business partners.

KEYWORDS
cloud-based ERP systems, digitalization, public sector management accountants, software-as-a-service, technology power loop

1 | INTRODUCTION

Digitalization increasingly permeates the private and public sectors, affecting all types of organizations and their practices (Hodges & Grubnic, 2010; Jeacle, 2021; Kinder, 2010; Moll & Yigitbasioglu, 2019). Regarding public administration, researchers and practitioners have articulated a new paradigm of "digital government," which promises more integrated, agile, and holistic public sector organizations (Andrews, 2019; Gil-Garcia et al., 2018). One technology frequently mentioned in this context is cloud-based computing (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Yau-Yeung et al., 2020). Cloud-based computing has increased in popularity due to its promise of enhanced flexibility and a lower cost of deployment (Moll & Yigitbasioglu, 2019). This technology entails a large part of an organization's IT infrastructure being outsourced to an external vendor or service provider. Basically, cloud-based computing enables actors to access applications and data on-demand from anywhere in the world at any time (Buyya et al., 2010). This allows a convergence of IT efficiency and business agility (Marston et al., 2011), with the possibility of rapidly scaling a system up (or down) based on a "pay-as-you-go" model. Given this promissory discourse surrounding digital technologies, it can be expected that—in the spirit of advancing digitalization of the public sector—the use of cloud-delivered systems, such as cloud-based enterprise resource planning (CERP) systems, will become more common in the future. As will be illustrated below, some public sector organizations are already using CERP systems and gaining experience with them.

In this paper, we explore how the implementation of a CERP system in a public sector organization affects its management accounting function. Although research has discussed CERP systems from a conceptual perspective, in-depth empirical research on whether (and, if so, how) this technology affects the role of public sector management accountants is lacking. Prior research has demonstrated that, while not a straightforward endeavor, public sector management accountants often have an ambition to establish themselves as business partners (Holmgren Caicedo et al., 2018; Järvinen, 2009; Mack & Goretzki, 2017; Moll & Humphrey, 2007; Paulsson, 2012). Cloud-based solutions appear to be capable of facilitating this aspiration as they are often presented as having the potential to unburden management accountants from transactional activities, thereby freeing up resources for more "value-adding" activities. Moll and Yigitbasioglu (2019, p. 4), for example, state that "unlike an on-premise ERP system, cloud delivery should allow accountants to spend less time backing up their data." Becker and Heinzelmann (2017) further mention that information produced through cloud-based technologies requires management accountants' skills verifying its reliability, reasoning that this might create "a major opportunity for management accountants to deepen and stabilize their aspired role of business partner" (p. 226).

Although cloud-based technologies might be regarded as potentially enabling for public sector management accountants, empirical research cautions us not to leap to conclusions. Empirical research draws an ambiguous picture of how technologies driving standardization in organizations impact the role of management accountants. Some

authors find that these technologies can facilitate a business partner role by relieving management accountants of routine activities (e.g., Goretzki et al., 2013; Järvenpää, 2007; Scapens & Jazayeri, 2003). Others, however, have demonstrated that they may lead to ambiguity around management accountants' roles (e.g., Caglio, 2003), even triggering tensions within the organization that hinder business partnering (Dechow & Mouritsen, 2005; Goretzki et al., 2018; Heinzelmann, 2017; Jack & Kholeif, 2008). Taken together, prior research shows that management accounting is not necessarily IT-enabled (cf., Becker & Heinzelmann, 2017; Lindvall & Iveroth, 2011) but, at times, may be an IT-constrained occupation. Thus, despite the current "hype" surrounding digital technologies, one needs to be careful when assuming that CERP systems are (unconditionally) enabling for public sector management accountants and their role performance. Rather, a nuanced analysis is required to evaluate how the promissory discourse of cloud-based systems materializes in practice and how this affects management accountants working in public sector organizations.

Responding to calls to explore the effects of cloud-based solutions on management accountants (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019; Yau-Yeung et al., 2020), we conducted a single case study of a large Swedish public sector organization (referred to as PubliCo). Prior research teaches us that to understand how technologies affect the roles of management accountants, one needs to pay attention to the dynamics around the design and use of those systems—as well as the actors involved (see, e.g., Ezzamel & Burns, 2005; Goretzki et al., 2018; Vaivio, 1999). Correspondingly, a research perspective that melds the technical and the social is appropriate. Accordingly, we draw on the technology power loop model (Scarbrough & Corbett, 1992) directing our attention to: how the CERP system is developed; who "controls" the system, and ultimately decides on the features to include; the ideology embedded in the system; the kinds of expertise inscribed in the system; and how this, in turn, enables or constrains public sector management accountants to mobilize their specific expertise in performing their roles.

Our findings suggest that cloud-based technology assumes an ambivalent role in the context of the work of public sector management accountants. On the one hand, the CERP system enabled central management accountants to mobilize their specific expertise by eliminating manual work, increasing transparency, and making central management accountants feel more comfortable with the numbers. On the other hand, the CERP system constrained local management accountants trying to mobilize their local expertise through the production of "tailor-made" information for their managers. In contrast to on-premise ERP systems, cloud-based systems are not only standardized and centralized but also externally managed, which makes it more difficult for local management accountants to inscribe their expertise into these systems and influence their design. We detail how the features of cloud-based technologies, such as limited upfront customization and outsourcing, posed a challenge for PubliCo's local management accountants and limited their ambitions to act as business partners. The case of PubliCo also illustrates that some of the features of cloud-based technologies (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019; Yau-Yeung et al., 2020) can indeed be beneficial for central management accountants—while creating challenges and problems for local management accountants. As such, our case adds to the discussion about obstacles to public sector management accountants' endeavors to establish a business partner role (see Holmgren Caicedo et al., 2018; Mack & Goretzki, 2017; Moll & Humphrey, 2007; Paulsson, 2012), while also emphasizing the importance of a nuanced perspective when studying the effects of cloud-based technologies on the finance function in public sector organizations.

The remainder of this paper is structured as follows. Section 2 presents the relevant literature and conceptual background for our study. The research methodology is described in Section 3, followed by our empirical findings in Section 4. The final sections provide a discussion and conclusion.

2 | CONCEPTUAL DEVELOPMENT

2.1 | Cloud-based enterprise resource planning systems

The phenomenon commonly known as cloud-based computing is argued to change the way information technology is delivered, deployed, used, maintained, and resourced (Marston et al., 2011; Yau-Yeung et al., 2020). Moreover,

TABLE 1 Main differences between traditional ERP systems and CERP systems

System specifications	ERP	CERP
Location/development	*On-premise*	*Vendor*
Customization	*Upfront and continuing*	*Limited*
IT service	*In-house*	*Outsourced*
System management	*In-house*	*Outsourced*
Scalability	*Moderate/High*	*High*

cloud-based computing is proclaimed to be a key digital trend for both private and public sector organizations informing their strategies (Bharadwaj et al., 2013). The cloud is presented as an efficient and adaptive IT infrastructure that can be accessed and shared on-demand among different end-users through web applications (Buyya et al., 2010), whereas a third party hosts the software on a remote server storing and processing the data. So-called software-as-a-service (SaaS) is an increasingly popular cloud-based alternative to the traditional on-premises software setup (Feng et al., 2018). It is a software licensing and delivery method that enables organizations to reduce physical hardware storage and to outsource IT management and services. It is a service model whereby the consumer can use the provided applications running on cloud technology, without having to manage the underlying infrastructure (NIST, 2011). Cloud-related services are generally described as being more economical for organizations due to a flexible provision of services and other cost efficiencies (Huang et al., 2015). With SaaS, the user no longer maintains the physical servers or cloud-based application software, instead paying a subscription for using the provider's already developed applications while accepting limited possibilities for customization. An organization's shift toward SaaS thus entails giving up some control over its IT infrastructure (Buyya et al., 2010).

Cloud-based ERP systems (CERP) are based on both cloud-based computing and SaaS, providing organizations with new opportunities for efficiency and agility in managing their IT demands. Like traditional ERP systems, they include multiple organizational functions such as accounting, procurement, and production; however, the software is no longer stored or developed in-house. Bryson and Sullivan (2003), for example, argue that these systems result in more efficient operations and the ability to manage cash flows more effectively by having predetermined monthly IT costs. The organization can focus on its core operations and does not need to recruit and retain expensive IT personnel. Olson (2007) outlines further benefits in terms of a reduction in capital expenditure for software and updates, continuous access to the latest technology, and more flexible and agile IT capability. Table 1 summarizes the main differences between traditional ERP systems and CERP systems.

Despite the promissory discourse associated with cloud technology, research on how this technology affects organizations and their finance functions is relatively scarce. Empirical examination of this phenomenon is crucial because, despite its promised advantages, cloud technology "produces significant management challenges, including maintaining strategic control, managing cloud services and dealing with the organizational, cultural and integration challenges" (Bhimani & Willcocks, 2014, p. 485). Several scholars have highlighted the need for research that adds to our understanding of whether and how cloud-based solutions affect the role of management accountants (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019; Yau-Yeung et al., 2020).

2.2 | CERP systems and the role of the public sector management accountant

The above-mentioned developments surrounding digital technologies present an interesting area for research on public sector management accountants who have long been known to embrace a varying set of tasks (Holmgren Caicedo et al., 2018; Mack & Goretzki, 2017; Moll & Humphrey, 2007; Paulsson, 2012). Their traditional role was to deliver routine financial analysis, reports, and information, which has been characterized of less relevance for decision-making

and strategy formulation. Such a bean counter role (Friedman & Lyne, 1997; Granlund & Lukka, 1998; Järvinen, 2009) focuses mainly on the provision of backward-looking financial information. However, as part of the new public management (NPM) movement (Paulsson, 2012), public sector management accountants have developed more enlarged ambitions to adopt increasingly strategic and advisory roles (Holmgren Caicedo et al., 2018; Mack & Goretzki, 2017). Such a business partner role is oriented toward interdisciplinary activities and linked to accountants' interactions with, and interventions in, managerial work in a more forward-looking manner (Burns & Baldvinsdottir, 2005; Goretzki & Messner, 2019; Järvenpää, 2007; Järvinen, 2009).

Although the bean counter and business partner roles are often presented in a dichotomous manner, recent studies within the public sector have problematized their unduly strict distinction (Holmgren Caicedo et al., 2018 ; Mack & Goretzki, 2017; Paulsson, 2012). Paulsson (2012 p. 391), for instance, criticized this "classic" typology by asserting that "it is difficult to find one single label like bean counter or business partner that fits the role of the management accountant." Mack and Goretzki (2017) illustrated this point by demonstrating how management accountants often occupy an in-between position and do not enact one stereotype only. Looking at the promissory discourse on cloud-based technologies and the complex role public sector management accountants occupy, the following research question arises: *How do CERP systems enable the role performance of public sector management accountants?* Addressing this question allows us to mobilize a public sector context to answer recent calls for research on the effects of cloud-based technologies on management accountants.

Overall, our study is situated in a broader domain that looks at the relationship between new technology and the role of management accountants. This domain has produced a somewhat ambiguous picture of the effects of technology on management accountants, especially concerning their ambitions to be(come) business partners (Becker & Heinzelmann, 2017). Some studies argue that ERP systems can facilitate a business partner role because they release management accountants from routine activities and grant them more time for business analysis (Byrne & Pierce, 2007; Sánchez-Rodríguez & Spraakman, 2012; Scapens & Jazayeri, 2003). Also, if management accountants interact closely with managers throughout the implementation process, they can build closer relationships and develop a better understanding of the business and managers' information needs (Caglio, 2003; Goretzki et al., 2013). The fact that ERP systems drive standardization and integration has been presented as a facilitating factor for the management accountant. ERP systems increase transparency and allow accountants to see what is going on in the different parts of the organization, enabling them to better control activities (Quattrone & Hopper, 2005). Other studies, in contrast, argue that ERP systems do not necessarily create positive effects for management accountants. For instance, the implementation of an ERP system may support operational managers by making them more independent of the finance function. This relates to the IT-enabled diffusion of accounting expertise throughout the organization, whereby managers start working with accounting information without involving accountants (Holmgren Caicedo et al., 2018; Hyvönen et al., 2009; Kurunmäki, 2004).

Further, prior studies have also outlined how the design and functioning of an ERP system impacts the role of the management accountant. Jack and Kholeif (2008) illustrated how the introduction of an ERP system was associated with the reversion of the management accountant's role to more bean counter types of activities. Heinzelmann (2018) makes similar observations and argues that "the implementation of IT systems does not necessarily steer management accountants' roles towards more management support and analysis," rather sometimes "more routine work and reporting ('dirty work') emerges from the extensive use of IT systems" (p. 479). In a further article, Heinzelmann (2017) explores how an ERP system's conceptual underpinnings affect the management accountant. Examining how accountants in a multinational context experience SAP R/3, he shows that since the system draws on a "German accounting logic," its implementation can create challenges for management accountants in non-German speaking countries who often must adjust the way they work to the specifics of the system.

Heinzelmann's (2017) study highlights the importance of considering the ways and the extent to which (cloud-based) technology affects the (public sector) management accountant. What expertise is inscribed in a system and how does that define the types of information produced (cf., Ezzamel & Burns, 2005; Vaivio, 1999)? Prior research has noted that local management accountants feel that standardized systems do not grant them sufficient flexibility to

produce information that can be used for business partnering. Goretzki et al. (2018) showed how local management accountants rely heavily on locally developed and maintained Excel spreadsheets to produce information that better supports local managers than information produced through standard systems. Thus, despite the promise that standard ERP systems can free up resources, local management accountants have chosen to develop additional resources because they prefer to work with "vernacular accounting systems" (see also Carlsson-Wall et al., 2019; Kilfoyle et al., 2013) that are locally meaningful.

Accordingly, CERP systems are a double-edged sword for public sector management accountants—enabling *and* constraining their role. Hence, it is crucial to acknowledge that the implementation of a CERP system is not merely a technical process. It also involves various expertise groups interacting and trying to make the system work for their purposes (Dechow & Mouritsen, 2005; Wagner et al., 2011). Thus, to understand how CERP technology affects public sector management accountants one needs to understand how these systems are developed, the nature of the expertise inscribed in them, and how, in turn, this enables or constrains actors to mobilize their expertise. The so-called technology power loop model (Scarbrough & Corbett, 1992) is a suitable analytical tool for understanding this and will be explained in the following section.

2.3 | The technology power loop model

The technology power loop model was developed by Scarbrough and Corbett (1992) and highlights how technologies are made to work, as well as how the role of expert groups may change as they seek to influence their development. Accordingly, this model conceptualizes a dynamic interplay between technology and power. In this regard, it directs our attention to how a technology is *developed;* who "*controls"* its development and decides on the features to be included in its development; the kind of *ideology* underlying the technology; and the types of *expertise* inscribed in it (Scarbrough & Corbett, 1992). Expertise is of relevance because actors leading the process of technology development draw on their specific expertise to (re-)design a system, shaping which group(s) of actors control a technology. Ultimately, this will also affect the extent to which a technology enables and/or constrains the expertise of a range of actors connected to it, including the extent to which technology may also (re-)define their expertise. The model allows for an examination of the potential tensions between different groups (both internal and external to the organization) involved in the development of the technology and vying for control over it. Thus, and of relevance to our study of the implications of CERP systems for public sector management accountants, "[t]his model [thus] illustrates a dynamic conception of expertise and a central role for technologies in supporting [or not supporting, certain] forms of expertise" (Newman & Westrup, 2005, p. 261).

Newman and Westrup (2005) used this model to explain how an ERP system was made to work and how different internal and external groups sought to influence its development. Considering the development of an ERP system as a contested arena, they argued that management accountants are (just) one group among many and it is not given that they will come out "on top" in such contests. The destruction of the monopoly of the finance function in the development of accounting information systems (AISs) can thereby shift expertise from management accountants to senior management and/or other groups, such as software vendors. The authors concluded that the success of an ERP system is not a purely technical issue, but rather depends on how both internal and external stakeholders interact with a system (and each other). However, they also argued that while accountants might cede some of their control of the system to other groups, they "collectively and individually make ERP systems work as they reposition themselves from being providers of information to becoming information analysts" and that this is another "instance of where the skills of accountants are needed to make the best use of these technologies" (Newman & Westrup, 2005, p. 269).

The technology power loop model provides an interesting conceptual frame for our study. CERP systems have been discussed as potential facilitators for business partnering by unburdening management accountants of routine activities (e.g., Becker & Heinzelmann, 2017; Moll & Yigitbasioglu, 2019). Additionally, these systems are argued to advance the provision of decision-making information (Strauss et al., 2015), which improves the information base and support

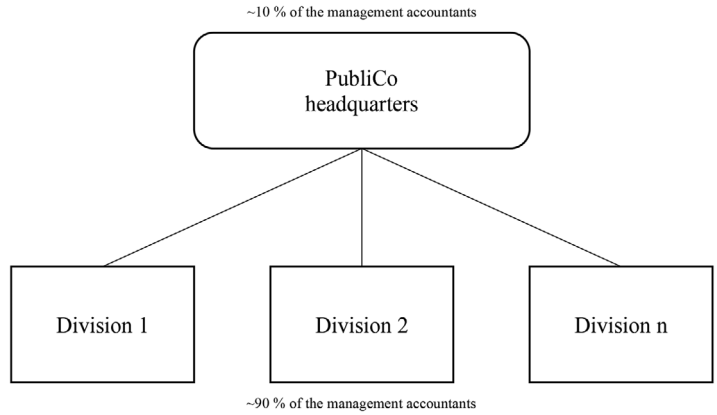


FIGURE 1 Organizational structure and the location of management accountants in PubliCo

that management accountants can offer their managers. Commentators have further argued that as IT expertise is no longer needed in-house, system development can be realized faster. CERP can present a digital leap forward in a public sector organization's technology because it introduces automatic upgrades, improves accessibility, and moves the system from the organization to the vendor. In terms of the technology power loop model, however, such outsourcing has the potential to create challenges for management accountants as vendors have a degree of control over the system and its development (Scarbrough & Corbett, 1992). Making a CERP system work—for the organization and management accountants as a specific occupational group—is more than an issue of the system's (technical) design elements and in-house (technical) expertise. It is an ongoing dynamic interaction between the CERP system and various groups of actors, such as management accountants at different organizational levels (Goretzki et al., 2018; Heinzelmann, 2017), in-house system developers, vendors, and service providers.

Pertinently, Scarbrough and Corbett (1992) note that the development of a technology is not a neutral process—but is always shaped by certain ideologies influencing the meaning(s) that actors attach to technology and reproduce through its design and implementation. The authors argue that "[i]deology filters and shapes the way in which people perceive, understand and relate to a given technology, and it is the meanings attached to technology that define the technically possible and the technically desirable" (p. 74). They also argue that "ideologies embodied in technology not only serve to define the 'legitimate' role and status of the user, but also the role and status of those who control the technology" (p. 75). The purposive introduction of a CERP system with its underlying premises, promises, and potential constraints reflects a certain ideological view of the potential role of technology within an organization and for various related groups of actors. Mobilizing the technology power loop model as our analytical tool thus directs our attention to whether and how public sector management accountants use a CERP system to perform or even expand their roles, as well as the ways and extent to which other groups of actors can take control of and benefit from these systems. How this played out in practice in the context of our case study of PubliCo is elaborated below.

3 | RESEARCH METHODS

To capture the complexity of a CERP system implementation, we draw on a single case study (Ahrens & Dent, 1998; Lukka & Modell, 2010). We do so to unpack the effects of how cloud-based systems are used in practice and how they are implicated in the performance of management accountants' role. The case organization—referred to as PubliCo throughout the paper to preserve anonymity—is a large, internally diverse, Swedish municipality. It is composed of multiple divisions including city planning, culture, labor, sports, education, environment, elderly care, housing, traffic, real estate, waste management, and parking. Each division functions like a stand-alone organization with divisional managers and a local finance function. As illustrated in Figure 1 below, most of the management accountants work in

the divisions (hereafter called "local management accountants"), with only 10% working at the headquarters (hereafter called "central management accountants").[1]

We knew from personal contacts that PubliCo was implementing a CERP system and these contacts facilitated our access to the field in 2018. PubliCo implemented the CERP system in late 2015 in two different phases and we managed to get access to divisions from both phases. When our study commenced, early adopting divisions (Division1 and Division 2) had worked with the system for two and a half years and the other divisions (Division 3, Division 4, and Division 5) had only completed the implementation less than a year ago. On entering the field, our general aim was to "track" the effects of this implementation on management accountants "in action." However, the implementation of this new system turned out to be a sensitive issue (Ahrens & Dent, 1998). Correspondingly, maintaining access was challenging as interviewees were either reluctant to talk about the change process or simply too busy (partly because of the new CERP system). Despite this, we managed to anchor our project at multiple levels in the organization and were able to interview four central and 12 local management accountants, four central directors, and two central systems designers. In total, 20 interviews were conducted with 22 different interviewees during the spring of 2018 (see the Appendix for full interview details).[2] It was important to interview both central and local level management accountants to understand if they were affected similarly or differently by the new system. The directors and system developers from the headquarters provided us with details about the ideas behind the implementation of the CERP system, as well as its design and how the organization "functions" in general. The semistructured interviews were conducted on-site and face-to-face, lasting between 40 and 75 min. All interviews were audio recorded and transcribed. We asked questions about the interviewees' daily work, routines, processes, relationships, as well as communication around and experiences with the CERP system (including a comparison to the old system). We also focused on the implications of this for the management accountants. In addition to our interviews, we reviewed both internal and publicly available documents.

In line with an abductive research approach (Lukka & Modell, 2010), data collection and analysis were highly integrated and proceeded iteratively. It became apparent throughout the research process that the technology power loop model (Scarbrough & Corbett, 1992) constituted a fruitful sensitizing device to analytically capture the nuances of the CERP implementation. In qualitative research, it is important to remain open to emerging issues (Lukka & Modell, 2010). In our case, this mainly related to the emergence of differences between how the implications of the new system were perceived by central and local management accountants and their varying attitudes toward it. Although previous literature had described how the implementation of a standardized and integrated system can change the role of management accountants within an organization toward either increased business partnering (Järvenpää, 2007; Goretzki et al., 2013; Scapens & Jazaveri, 2003) or bean counting (Jack & Kholeif, 2008; Heinzelmann, 2018), we found both consequences within PubliCo.

All researchers were involved in the data analysis, wherein the empirical material was first arranged in terms of common themes, such as management accountants' roles (e.g., tasks, processes, and relationships), and specifics linked to cloud technology (e.g., external system providers, limited customization). The narratives resulting from these parts of the analysis are presented in Section 4.1 and Section 4.2. We then related our data to the main theoretical concepts of the technology power loop model (i.e., how the CERP system is developed, who "controls" it, what ideology is underlying it, what kind of expertise is inscribed in it, and how this, in turn, affects the management accountants). This part of our analysis, reported in Section 4.3, detailed the *development* and *control* of the CERP system. We observed that responsibility for technological development became increasingly centralized and outsourced. Drawing on the technology power loop model, we interpreted this as resulting from a shift in the *ideology* underlying system development at PubliCo. Theorizing our data, we detailed the kind of *expertise* that was inscribed in the CERP system and how this, first, enabled central management accountants' expertise (see Section 4.4) and, second, constrained local management accountants' expertise (see Section 4.5). A further element of our analysis (see Section 4.6) examined the consequences of the increasingly centralized and outsourced development and control of the CERP system on local management accountants' mobilization of their specific expertise in performing their role. As such, the technology power loop model was pertinent in guiding the analysis of the empirical data. Through organizing and

structuring the empirical material into quote tables based on the categories deduced from the technology power loop model (described above) and categories developed inductively from our data (i.e., the need to differentiate between central and local management accountants), we built a nuanced account with multiple "voices" (Lukka & Modell, 2010). This allowed us to deepen our discussion and develop a theoretically guided narrative that enriches our understanding of how cloud-based technologies affect public sector management accountants, including, more generally, how digital technology affects management accountants in multi-layered organizational structures.

With respect to generalizability, it is noteworthy that statistical generalization is not the primary objective in a qualitative case study (Gobo, 2008; Scapens, 1990). The intention is to make sense of the case at hand, to get to the bottom of a specific problem, concern, or question and develop fine-grained knowledge about it. Qualitative case studies in this sense draw on the notion of *analytical generalization* and do so in two ways (Polit & Tanano Beck, 2010). First, when articulated in a manner that is authentic and credible to the reader, abstracted explanations for specific processes, actions, or events represent theorizations that can be transferred to other cases. Subsequent studies can then apply these explanations to other cases to further extend the explanations—for example, to different organizational contexts. If the theory developed from a previous case provides explanations of the new cases, it could be said to be (analytically) generalized, as it not only explains a specific case but also other and possibly more diverse cases. The explanations thus have a theoretical resonance that goes beyond mere empirical description. Second, explanations derived from a particular case could be extended to other cases because they have similar theoretical characteristics. In this way, theoretical arguments are used to apply the explanations of this case to a broader set of phenomena.

4 | EMPIRICAL FINDINGS

4.1 | Introducing PubliCo and its management accountants

PubliCo's headquarters are responsible for governing and coordinating municipal planning, IT development, administration, and human resources. It serves a management function and has the responsibility to ensure viable short- and long-term financing through active financial governance. The central finance function is responsible for drafting the budget proposal (including all divisional budgets) for approval by municipal politicians. A central IT team, which is part of the municipality's finance function, is responsible for all central accounting and financial IT systems (including the implementation of the CERP system).

PubliCo has a long-term vision that embraces four overall objectives, with these encompassing 26 central level strategic goals related to, for example, equality and antidiscrimination, safety, climate and environment, sustainability, innovation, digitalization, and availability. These strategic goals are imposed on every division, which means that some or even all divisions could be jointly responsible for achieving a specific strategic goal.

PubliCo's most important financial target is to have a balanced budget. However, when this goal is fulfilled, the focus shifts to evaluating and assessing the value PubliCo has provided for its stakeholders through the fulfilment of the 26 strategic goals. These assessments are publicly available for stakeholders. As all operational activities are performed by the divisions, they are ultimately responsible for achieving the strategic goals. To properly exercise their responsibility, the divisions usually break the strategic goals down into more concrete divisional goals.

Both central and local municipal management accountants perform bean counter oriented tasks related to performance evaluation, reporting, and financial follow-ups, as well as business partner oriented activities related to decision support, analyses, and the management of strategic issues. More specifically, the local management accountants' bean counter oriented tasks consist of providing financial information to headquarters for financial follow-up and production of financial statements. This information helps top management evaluate divisional performance. The central management accountants' bean counter oriented tasks relate to the financial follow-up of the divisions and the preparation of the financial reports for the whole municipality. As PubliCo is highly decentralized, central management accountants usually do not get involved in the divisional operations.

One of the most important business partner oriented tasks for both central and local management accountants involves designing and working with performance measures. For the central management accountants, this involves forecasting and assessing the degree of fulfilment of the strategic goals and presenting their assessments to the top management team. These assessments are based largely on the outcomes of underlying activities and indicators. However, they are often asked to propose specific performance measures for different goals. Central management accountants stressed how important it is for them to find ways to free up time for these activities by engaging less in tasks related to compiling the financial reports and scrutinizing the quality of the financial data and reports submitted by the divisions.

When the local management accountants perform tasks that they consider business oriented, they are supporting divisional managers in their operational activities. As the local management accountants are familiar with both central as well as divisional goals, divisional managers see them as key resources to fulfil their strategic goals. Local management accountants mentioned that the divisional managers are not overly familiar with the financial consequences of their decisions. Consequently, they need to "educate" the managers from an accounting perspective, acting as a "devil's advocate" when it comes to the budgeting process and supporting them in the 3-year strategic planning process.

4.2 | The new CERP system within PubliCo

The decision to implement a CERP system was taken by the municipal council following a long period of consultation whereby individuals from different divisions were invited to propose specifications to be included in the public procurement documents. As the divisions worked with a variety of on-premise local AISs, the aspiration was that the new CERP system should make reporting faster and simpler, increase cost efficiency by decreasing administrative work, and simplify future system development. Further, it should also enable the management accountants to focus on "value-creating" tasks:

> We hope to free up time for the management accountants in the divisions, time they can allocate to analyses and support. The idea is not to save money [by laying off management accountants], it is more that we want better quality. The time they save should be used to create value for the divisions. (Headquarters Finance Director 3)

Vendors were then invited to bid on the project if they fulfilled the regulatory requirements mandated by PubliCo. Several bids came in and the central IT team decided on a vendor. The chosen CERP system includes several modules (procurement, accounting, inventory, project management, and billing) and serves a unified purpose of cloud integration. Additional systems, such as local customer relationship management (CRM) systems, were linked or integrated to the CERP system. The modules have a standard format, requiring the adaption of all divisions to them. The CERP system thus replaced the different local AISs for procurement, accounting, inventory, and so on, which were used in the respective divisions of the municipality.

The resultant contract was drafted to include two external parties: the application system vendor (ASV) and the service provider (SP). The ASV is responsible for developing and updating the system and the SP oversees server maintenance, technical support, testing, and implementations. The users access the system through the cloud via a web application and, in the event of a problem, can contact the SP for support. The communication between the ASV and PubliCo is conducted solely via the central IT department and focuses on updates and development requests. The CERP system was implemented gradually in the divisions so as not to overburden the central IT team. The setup and communication flow around the system is illustrated in Figure 2 below.

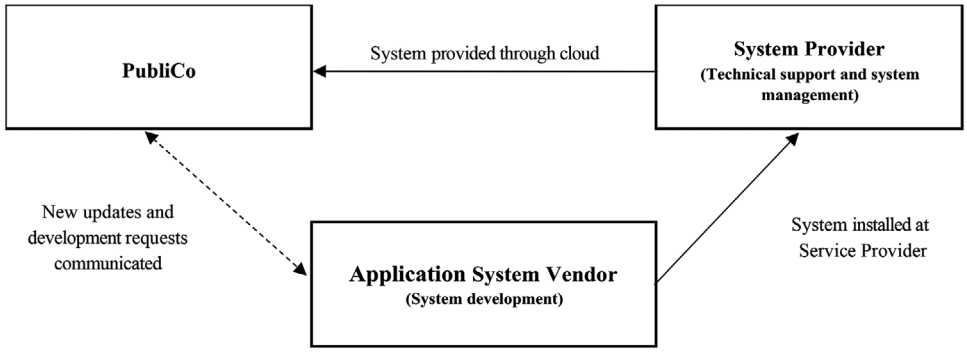


FIGURE 2 Illustration of the system setup and communication flow for the CERP within PubliCo

4.3 | The development and control of the CERP system

A crucial feature of the CERP system was the centralization of the IT development process, creating a standardized way of working throughout the entire organization. This led to a major change in PubliCo as to how systems were developed and who controlled technology (Scarbrough & Corbett, 1992). Local management accountants described the pre-CERP era as a "pain-free process": they would work together with the local IT team to develop local on-premise AISs that catered for their local needs, allowing them to provide divisional managers with the information required to manage their strategic goals. With the introduction of the CERP system, however, the responsibility for the system shifted to the central IT department and (partly) to the external ASV and SP as well. The central IT department became responsible for managing the implementation process and coordinating with the external partners:

> We do not do any coding and programming [ourselves]. We are responsible for the processes. If there is something with the code, then [the ASV] is doing that. We have an opinion about what processes and modules work or not and how we want to change the system. Then maybe the [ASV] says it is not possible in the standard solution, then we decide if we need an adaptation. If so, we tell them what to develop and they do the programming and coding. (Headquarters System Developer 2)

Although the decentralized divisions were used to having a free rein to develop their local systems, with the CERP system every modification to the off-the-shelf solution had to be evaluated and approved by the central IT department. Throughout the implementation of the new system, the central IT team thus assumed a stronger position in the organization. It established itself as a driver of standardization, efficiency, and technological advancements, approving only a few of the local change requests:

> … we are the 'owners' of the process. If there is a problem, we are responsible. The [SP or the ASV] would never do any major things without talking to us. And we do not authorize any changes if there is not a major thing of importance for the whole municipality. (Headquarters System Developer 1)

This was seen as a relatively radical step. Prior to the CERP system implementation, the divisions controlled their systems and could develop and adjust them at will. The local management accountants thus felt that their role had been circumscribed—while the central IT team had improved its position. System Developer 1, who had acted as the project leader for the CERP system implementation, explained how the IT department had grown and occupied a more strategic role within PubliCo:

> We [the central IT team] have two responsibilities. The first is making sure the system works. We purchase that service externally. The users call our service provider if they need support. Our second focus, our main focus, is strategic. What should we do? How should we do it? What projects should we go for? What are the effects? What choices should we make?

Connecting our observations to the technology power loop model (Scarbrough & Corbett, 1992) and especially the link between the development and control of technology, despite PubliCo's highly decentralized structure, the responsibility for technological development became increasingly centralized and outsourced. The central IT department took control of the new system, as will be further elaborated in Section 4.6. In contrast to Newman and Westrup (2005), however, the central IT department's position did not result from a battle between different occupations. Rather, drawing on Scarbrough and Corbett (1992), we can interpret this as resulting from a shift in the ideology underlying system development at PubliCo. Centralizing development was seen as crucial to reducing activities within the central and local finance functions that were "non-value adding." We can interpret this as an ideological change as to how technology empowers management accountants to perform a business partner role. In the pre-CERP era, the belief was that this could be achieved through highly localized (on-premise) AISs, wherein management accountants (in conjunction with local IT) were both key producers and controllers of the systems. The ideology underlying the CERP project, however, was that standardized and integrated systems developed and maintained externally would allow management accountants to "outsource" a large part of their bean counter activities and, thereby, concentrate on using ready-made information for more detailed analysis. Consequently, the control of the system by the central IT department was a by-product of a new "techno-ideology," rather than the result of an occupational competition between the local and central management accountants.

In the following section, we elaborate how local systems replaced by the CERP system affected the work of central and local management accountants.

4.4 | Central management accountants' perceptions of the CERP system—An enabler for centralized expertise

In our case, centrally located management accountants developed a positive attitude towards the CERP system and acknowledged that it helped them perform their tasks. Prior to CERP, they had to manually transfer data from the local systems into the group reporting tool, which they perceived not only as a significant time drain but also a source of potential errors. The new system minimized this risk of errors by automating information flows and reducing the need for manual data input:

> [Before the CERP implementation] it was the classical, often with Excel spreadsheets … But now they upload the data in [the CERP system] so we see it directly … Much easier for us since we do not need to adjust and cut and paste the data from Excel to our system. […] It is better and safer [now] … Nothing should be wrong. It is the taxpayers' money so there should be no errors. (Headquarters Management Accountant 1)

For them, the CERP system had improved data quality and made them feel more comfortable with the numbers:

> The [CERP system] is where the balance sheet and income statement are, that is the closing of the books. It needs to be correct here … One should not sit with various files and Excel. It is not safe … the new system is much safer and better. (Headquarters Management Accountant 2)

This view was also reinforced by the finance directors:

> [With CERP] it is a more stable system, more stable numbers … before we had to sit with different Excel files, with various links. This almost created a gastric ulcer, 'have we changed the formulas correctly?'… all numbers [from the divisions] had to be saved manually and worked with manually since they were in different formats … now everything is automated … all these thousands of links etc. that we had to deal with before, such a big error source … so we are much more secure today … and it is the same tasks we do. It just goes so much faster. It was so much [manual] work before to go through all the various files. It took months to see that it was all correct. […] Now we can get all the data we need much faster and safer. (Headquarters Finance Director 3)

Both the finance directors and central management accountants emphasized that as increased transparency and automation facilitated less record-keeping activities and increased comfort with the numbers, the central management accountants could focus more on business-related matters rather than manual bean counter activities:

> Of course, when the financial numbers are transparent and secure, they [central management accountants] can focus more on the strategic goals and analyses, preparation and follow-up related to those goals. So, we have really freed up time for this type of work. "Yes, the budget [for a particular division] was 200 million. But have you done what you are supposed to do with these 200 million?" It is not only the financial goal [a balanced budget] that should be fulfilled, it is also the strategic goals. "We should build 2,000 apartments, we should employ x number of unemployed people", etc … now they can spend much more time on that since we can trust the [financial] numbers. (Headquarters Finance Director 1)

> Now we have more time to work on the strategic goals for the coming three years. A mix of very political goals and concrete goals. Some are politically decided, often quite visionary. Then they need to be broken down into more concrete goals … we often write drafts that need to be approved [by top management]. (Headquarters Management Accountant 3)

As outlined above, the beliefs underlying the CERP system materialized in the central management accountants' experiences of the system. For them, it made life easier and allowed them to use the information produced through the system for activities that they felt created more value for the organization. Like the central IT department, the new system indirectly facilitated their position in the organization. The fact that stricter standards helped the central management accountants to consolidate information in a faster and "safer" way, with this being recognized by the finance directors, led to an overall positive attitude at the headquarters level toward the centralized and partly outsourced approach to developing the system.

In Section 4.5 below, we will elaborate how this led to tensions among the local management accountants who felt that the new systems did not enable them to use their local expertise (see Vaivio, 2004) to perform a business partner role.

4.5 | Local management accountants' perceptions of the CERP system—A constraint on performing local expertise

When talking about the CERP system, many of the local management accountants acknowledged the advantages accruing to their headquarters. Interviewees, for example, mentioned that the new system would allow the central management accountants to create "nicer reports" and enable them to better carry out their specific tasks:

> A lot, as I see it, is about getting a common financial reporting structure. So, headquarters benefits if everyone reports in the same way. (Division 4 Management Accountant 1)

> I see also clear advantages with the templates which are the same for the whole municipality. So, if one of us [local management accountants] quits or changes jobs, there is a standard routine, and it is the same for everyone. (Division 2 Management Accountant 1)

Local management accountants also realized that this uniformity could even improve the communication among the divisions, with the pre-CERP era creating what Scarbrough and Corbett (1992) have called "islands of technology" constraining knowledge exchange:

> I think over time we talk much more to the other divisions and share knowledge now that we all are in the same system. I think that the various old systems hampered the cooperation [between divisions]. This is what I hope [will happen]. (Division 5 Management Accountant 2)

However, despite the advantages they attributed to the CERP systems, the local management accountants experienced significant issues, which (in their view) hindered mobilization of their local expertise in performing more "value-creating" tasks:

> A lot of manual work, this is a huge change for us. It takes more time. It is not good but there is nothing we can do … We now spend much less time on value creation, on the analytical work. Much less time for this since we are so into the manual data work, to get everything in the [CERP] system. (Division 4 Management Accountant 2)

> I really like to work with analysis and these types of things. Less time for this now. (Division 1 Management Accountant 3)

Their main concerns related to a perceived lack of flexibility associated with the new system and an increase in the manual work necessary to make it function. A key issue was that a standardized way of reporting, a uniform "master data" template, was created by the headquarters and each division had to report in a prescribed way. As a result, several previously individual accounts had to be consolidated into one. The purpose of this was to create more comprehensive accounts of the organization and to enable top management and central management accountants to effectively follow-up and evaluate goal achievement at a group level. However, major efforts were needed from the divisions to construct such uniformity across their data, as previous highly customized codes and processes had to be changed to accommodate municipality-wide uniformity. Some of the local management accountants considered this mandate intrusive and did not see any advantages from altering their previous (and, in their view, highly satisfactory) working practices. When asked how these changes had been communicated from the central office to the divisions, a local management accountant described his antipathy as follows:

> The unified account-plan was to be able to see things together. But there weren't many benefits for us [local management accountants]. I felt […] this wasn't for us but for someone else. (Division 4 Management Accountant 1)

Indeed, one of the motives of implementing the CERP system in the first place was to improve the reporting for the municipality *as a whole*, leading to increased transparency across PubliCo. Some local management accountants opposed this as they were used to being highly independent in terms of their daily operations and control of their local systems (cf., Scarbrough & Corbett, 1992). Hence, they perceived the ambition to increase organization-wide

transparency as invasive, especially as they felt that transparency had been reduced *for them* because they had access to a smaller amount of data.

Overall, local management accountants lamented that the new system was insufficient for value-adding operations within their divisions. This was mainly due to the central IT department's restrictive attitude, with many proposed critical local alterations being rejected. The most conspicuous illustration of the CERP system's lack of flexibility, and the consequences this caused for the local management accountants, is exemplified by an accounting practice of Division 4. Due to the complexity of its business model and a strong focus on long-term contracts, Division 4 had to recognize revenues and costs over periods stretching years into the future. The new system, however, only permitted accruals for 1 year, presenting a significant problem for the local management accountants to work around. They were now forced to put the contracts down as assets in another module and at every closing they had to manually extract the data from that module. It took several processes to get the right numbers and the task had gone from taking a couple of minutes to several days, even forcing the local management accountant to work weekends:

> With the new system many tasks are more long-winded, extremely so. It just takes so much more time … we work a lot with accruals, especially for revenues. This we did before in the old system in a very smooth way, it took a couple of seconds. Now we have to register this as a sort of non-current asset, so we have to perform a number of manual processes and put it in the system … it is really difficult to get it all in the system correctly. None of this I had to do before … [Central IT] says it cannot be done in another way in the [CERP] system, they cannot adapt the system to meet our needs, one simply cannot have multiple future periods open in the system. This is the reason for all this manual work … [Central IT] says it is unique for our division, and that is why they cannot modify the system just for us, all divisions should do it in the same way …. we called and yelled many times. But [they] did not give in at all …. This is many, many days of extra work for us, that did not exist [with the old system]. (Division 4 Management Accountant 2)

Other divisions experienced problems with the CERP system's invoicing module. For example, Division 5 is heavily reliant on customer data stored in a local CRM system. Their former legacy systems were specifically designed to communicate and transfer this data efficiently. To the local management accountants' frustration, the new CERP system was unable to interact with the local CRM system. Even after numerous discussions with the central IT team, the division was denied the necessary adjustments that they felt were needed to solve the problem. The resulting solution was to keep the old CRM system for preparing invoices and then manually transferring these into the CERP system, which took time local management accountants could otherwise have used to run analyses for their divisional managers. They consequently saw the system setup as "completely worthless." Interviewees complained:

> Handling of invoices did not work well, so to make it work we had to keep some parts of our old system. This is not good of course and a lot of extra work for us. (Division 5 Management Accountant 1)

> When it comes to the invoicing module for example, it is now much worse. It takes more time for us to handle the invoices, and this is now a big part of what we need to do. (Division 1 Management Accountant 3)

> Invoicing has been major problems in the [CERP] system. We have solved it with major manual work arounds. (Division 2 Management Accountant 3)

The local management accountants complained that despite the initial promise that the CERP system would reduce manual work, they found themselves in a situation where even more bean counting was necessary to make the new system work:

> It is much more manual work for us now. We have to manually re-book. Move data manually to get everything right … I understand that the reporting to headquarters is more efficient since everyone has the same system … but with our own systems we could modify things quickly. Now I have to go through [the SP] and often through central IT … before, we controlled the system ourselves, now [with the CERP] I am being controlled. And has one once been "free," and not anymore, this is difficult … I want to remove a row in the system. Very easy? Not anymore, it has to go through [the SP], fill in a form, I have to sign, a manager needs to sign, scan it, e-mail it. It becomes very slow and complex. (Division 2 Management Accountant 1)

Most local management accountants described negative experiences associated with the time-consuming manual work that emerged because of the new system not working in conformity with their local requirements. In their eyes, this canceled any efficiency gains the CERP might have created. Prior to the implementation of the CERP system, each division had highly adapted and locally developed systems servicing their business. These systems reflected the local management accountants' local expertise and enabled them to support their managers. By shifting to the CERP system, they saw themselves confronted with an off-the-shelf solution that forced them to adapt to a standardized format. Thus, local management accountants usually described their old systems in positive ways and indulged in fond memories of them, while adopting a negative attitude toward the CERP system:

> It has become tiresome … we actually had a very good system before that worked perfectly. Now there are several steps and several things to keep track of. Now everything needs to go through several systems. It has not at all become more flexible. (Division 4 Management Accountant 1)

> With the old system, it was perfectly adapted to our operations, we could say exactly what we wanted to have. But with the new system, we had to adjust to the system … this means much more manual work for us. (Division 5 Management Accountant 1)

Theoretically, the new CERP system imposed externally designed templates, routines, and ways of working that constrained the local management accountants. This frustrated local management accountants' ability to apply their expertise to produce information to support their divisional business managers' work (cf., Scarbrough & Corbett, 1992). They felt that the CERP systems embedded "foreign" beliefs about how things are supposed to work, constraining their ability to invoke "vernacular accounting" (Kilfoyle et al., 2013) practices that had proved their usefulness in previous situations advising and supporting their business managers. Given the local management accountants' negative attitude toward the CERP system, we discuss below the extent to which they were able to influence the system's (further) development (Scarbrough & Corbett, 1992).

4.6 | Local management accountants' issues with influencing the development of the CERP system

Due to the shifting "locus of control" for the system (Scarbrough & Corbett, 1992), many local management accountants saw their responsibilities circumscribed and their authority reduced. For example, the fact that communication with the ASV and SP was handled solely by the central IT team frustrated and limited them:

> Before, I had full access and I could just go into the system and change it. Now I don't have any authority to do that. (Division 2 Management Accountant 1)

They lamented that influencing the new system had become rather difficult and sometimes even impossible for them. This was especially the case as the central IT department adopted quite a restrictive approach when it came to allowing alterations to the system; only in exceptional cases were local change requests approved. When the CERP system was implemented, the system development process was divided into two categories: smaller and larger alterations. For smaller alterations, local management accountants could mobilize their local expertise and make requests directly to the SP and they perceived this process as smooth. When it came to requests for larger alterations, however, they experienced the process as rather cumbersome and frustrating:

> We cannot influence the development of the system [as much as before]. We can only ask for it from the central IT office. […] We used to have a lot more freedom in the old system … where we could develop it ourselves with that vendor, but now we must go through the corporate headquarters every time. (Division 1 Management Accountant 1)

The initial step was to contact the SP who evaluated the scope of the request. If they perceived it to be a larger development request, then the central IT office would undertake an internal evaluation to assess if the proposed changes were something only the requesting division needed or if the entire municipality would benefit from it. If the latter was the case, the request was approved; otherwise, it was rejected:

> Everything has to go through the [SP]. If there are big projects or requests, [the SP] contacts central IT. And then they [central IT] say: 'OK, is this something only [Division2] wants to have?', and then they say no. But if we request something that might benefit the whole municipality, then they might say yes. (Division 2 Management Accountant 1)

Consequently, it was not given that requests would be approved, which, at times, led to irritation among the local management accountants who were focused on supporting their division rather than the central level:

> We would of course rather have purchased our own [system] or kept developing our previous system. We were afraid we would not be able to influence the new system and forced into a solution that would not be optimal for us. Especially with the accounts payable ledger, where we had a very good and flexible solution [in the old system] … So, we do not really think about the [system] value for the whole municipality even if we of course understand the arguments. (Division 1 Management Accountant 3)

The fact that many requests were not approved was not the only problem. Another was the time it took for approved change requests to be implemented:

> Now that everyone has the same system, it has become such an enormous process. We are now part of a giant inflexible apparatus. You need to go through several levels, meetings, and managers … it has become so slow. It feels like the Soviet Union. It is not flexible at all. (Division 4 Management Accountant 2)

One interviewee who managed to get her request finally approved, lamented how her division's request had been on a priority list for 2 years without being resolved. The long procedure forced her and her colleagues to work with suboptimized processes, in effect reducing the time available for more "value-adding" tasks:

> We cannot control any of the development of [the CERP system]. We can only request and wish for modifications, but [central IT] decides. And they take into consideration all other divisions. It is a much slower process. With the old [local] system we were much freer and more flexible. We could then

> continuously develop the system together with local IT and the system provider. Now we have to go through [central IT] all the time. (Division 1 Management Accountant 2)

Overall, a feeling of powerlessness characterized the local management accountants as they experienced stricter limitations as to what they could do in the system (especially when compared to their old local systems):

> Instead of solving it ourselves, we now have to e-mail a request [to central IT and the SP], and then we get on a priority list. It can take months; half a year and we want it within three days. But given that it takes so long we sometimes do not even send in suggestions for modifications. (Division 2 Management Accountant 3)

> From a division's point of view, we think that the old system gave us a lot of freedom. Now with a centrally run system, we are forced into this. For good and bad. A common system is more secure, but if we want a particular modification, we do not have this freedom anymore … It was very clear to us that [central IT] did not want lots of modifications. It was decided early on to have as few modifications as possible … but all divisions are different with different needs. It is not possible to find a system that works for everyone. (Division 5 Management Accountant 1)

Several local management accountants explained that they had difficulties relinquishing their former tasks to the SP, pointing to the cumbersome process it now entailed:

> I'm happy with the system. It's working with it that's the problem … We used to have administrative system rights, and it's good we don't have to manage that anymore, but we can feel limited in the sense that we cannot see things. It's not possible, you have to ask the SP … I think being able to go in and look for myself would solve things faster and you wouldn't have to involve the SP. (Division 5 Management Accountant 1)

Our case illustrates how a cloud-based system with restricted opportunities for customization can be welcomed by central management accountants but create frustrations on the part of their local counterparts. Prior to the implementation of the CERP, local management accountants were involved in developing their local AISs and could inscribe their local expertise in those systems. However, they are now confronted with an off-the-shelf solution that only grants limited influence (Scarbrough & Corbett, 1992). Influence on the system is only possible if they have a compelling case in that their request is beneficial for the whole organization, contributing to making processes easier, more efficient, and safer within PubliCo. Considering this, the new system setup triggered power dynamics (Scarbrough & Corbett, 1992) that disciplined local management accountants in line with a centralized structure—even though the implementation of the CERP system was not part of a broader centralization strategy.

5 | DISCUSSION

The case of PubliCo demonstrates how the meanings and attitudes that central and local management accountants attach to a CERP system relate to the technology's ideological underpinnings, as well as their specific understandings of how technology can (or should) enable their work (Scarbrough & Corbett, 1992). This was especially noticeable when analyzing the shift within PubliCo with respect to how the role of the AIS "framed" the management accountants' work and systems development (Scarbrough & Corbett, 1992, p. 75). PubliCo's "old" system landscape resulted from a strong belief that local flexibility in the development of (on-premise) AIS would facilitate the work of the finance function. The introduction of the CERP system in this sense reflected different beliefs and assumptions about how

accountants work with information and how technology is supposed to empower them to cultivate a business partner role. Considering the overall ambition within PubliCo to produce "better, faster and safer" reports on all levels of the organization, the introduction of the CERP system reflected a shift from a focus on a local *production* of tailor-made information via on-premise systems toward a company-wide *consumption* of prefabricated information produced through a ready-made (standard) system developed and maintained externally.

Examining the introduction of a CERP system, our study showcases how ideological changes regarding the role of technology in the finance function shapes management accountants' ability to control AISs and how this affects how those systems enable or constrain accountants' expertise and role performance. We thus demonstrate how the introduction of a CERP system can trigger dynamics and tensions in what Scarbrough and Corbett (1992) refer to as an organization's technology power loop. Before the CERP project, local management accountants within PubliCo took part in the development of their own on-premise AISs. In terms of system development, they "controlled" those systems and were able to flexibly inscribe their divisional expertise as well as their managers' information needs into their systems (Scarbrough & Corbett, 1992). The local management accountants' position in the technology power loop was, in turn, reinforced through their self-developed systems as the latter mainly drew on and reproduced the accountants' codified expertise. Although this setup empowered the local management accountants and allowed them to perform a business partner role, it posed problems for their central counterparts. In the pre-CERP era, central management accountants were arguably detached from the (local) system development. The central management accountants were not involved in the system development, they had no control over the local systems that produced the numbers they were to work with, and they merely "consumed" the numbers their local counterparts delivered to them. Although their main issue was not that they were not in charge of the local systems, central management accountants often struggled with the consequences of the highly localized setups. For example, consolidating various types of local information and aligning different reporting formats efficiently and accurately was perceived as challenging and cumbersome. Central management accountants also felt that this strong reliance on the (old) systems based on the local management accountants' expertise further constrained them in demonstrating their respective expertise vis-à-vis their superiors. From their perspective, the (old) system did not allow them to properly act as central business partners who provided more advanced analyses for the finance directors and other senior managers. In contrast, they were required to act as bean counters—whose main task was to bring order into a messy information set so that local information could be used more confidently and meaningfully at the central level. The fact that the new CERP system provided numbers in a standardized format thus made it easier for the central management accountants to accomplish their desired role, providing top managers with "nicer" reports and devoting more time to analytical and "value-adding" tasks.

Although the purpose of the CERP system was to enable *both* local and central management accountants to pursue their work objectives, the central management accountants developed a more positive attitude toward it than their colleagues at the divisional level. The central management accountants felt that the shift toward standardization and integration triggered by the CERP system led to less manual work, reduced the risk of errors in the consolidated reports, and freed up resources so that they could perform more analytical tasks. The local management accountants, however, experienced the opposite situation. For them, a major issue was how the CERP system was developed and who controlled it (Scarbrough & Corbett, 1992). The management of the implementation process was assigned to PubliCo's central IT team, which designed the system in collaboration with external partners. This enabled the central IT team to effectively control the system (Scarbrough & Corbett, 1992). As customization of the CERP system was difficult, the central IT team approved local change requests only if the *whole* organization benefited from them. As a result, the local management accountants were no longer able to flexibly inscribe their expertise into the system (Scarbrough & Corbett, 1992). This was an issue for them because they were used to locally developing and controlling *their* systems and to producing *their* numbers, considering this "production" element as a crucial part of their business partner role. The local management accountants considered that the specific ideological underpinnings of the CERP forced them to hand control of the system to others, confronting instead a role as "mere" consumers of a product that did not satisfy their needs. They struggled with how the CERP system constrained them in performing what they saw as more

value creating tasks and how (in comparison to their central counterparts) *they* now had to perform extra manual work to make the new system work locally. Consequently, their business partner role was curtailed by the implementation of the CERP system.

Looking at the different attitudes that central and local management accountants developed toward the CERP system, we can see that although both focal groups in our analysis belong to the same occupation, they framed the role of technology in performing their expertise differently (Orlikowski & Gash, 1994; Scarbrough & Corbett, 1992). While local management accountants framed technology as a tool that should enable them to draw on their local expertise to produce tailor-made information for their local units, central management accountants viewed it as a tool allowing them to consume prefabricated, "high-quality" information to assure efficient and risk-free accounting processes throughout the entire organization. Given this, cloud-based solutions seem more enabling and (hence) appealing for centrally located rather than local management accountants. This helps to qualify the limits of the more general observation of Newman and Westrup (2005), who champion a viewpoint that technology provides an opportunity for accountants to shift "from being providers of information to becoming information analysts" (p. 269). In comparison, our case points to the fact that not all accountants experience this change in an equally enabling and transformative way.

It is important to emphasize that we could not identify a strategic agenda behind this shift that curtailed the work aspirations of the local management accountants (in the divisions) and improved that of their central counterparts (in the headquarters). As such, we did not observe an overt and increasing ambition to drive centralization within PubliCo. Rather, the sincere belief expressed by the finance directors was that management accountants at both the central and local level would benefit from the CERP system. By not having to develop and maintain the systems themselves and by performing less manual work related to the consolidation of the heterogeneous information sets, it was believed that business partnering would be facilitated within both the headquarters and divisions of PubliCo. While the central management accountants indeed benefited from the implementation of the CERP system, the local management accountants felt constrained and frustrated by it.

6 | CONCLUSION

This study provides insights into how the promissory discourse of cloud-based technologies materializes in the practices of the finance function in a public sector context. More specifically, it examines the experience of management accountants in a large, decentralized public sector organization with a newly introduced CERP system. Drawing on the technology power loop model (Scarbrough & Corbett, 1992) to understand a single case study of PubliCo, a large Swedish municipality, our study extends the literature on how technology affects management accountants, both in a public sector setting (Holmgren Caicedo et al., 2018; Mack & Goretzki, 2017; Paulsson, 2012) and more generally (e.g., Caglio, 2003; Goretzki et al., 2013; Heinzelmann, 2017, 2018; Jack & Kholeif, 2008; Newman & Westrup, 2005; Sánchez-Rodríguez & Spraakman, 2012; Scapens & Jazayeri, 2003). Building on the findings of previous research pointing to the ambivalence of the effects of technology on management accountants (e.g., Caglio, 2003; Dechow & Mouritsen, 2005; Goretzki et al., 2013; Heinzelmann, 2017, 2018; Jack & Kholeif, 2008; Lindvall & Iveroth, 2011; Scapens & Yazayri, 2003), we have been motivated to develop a more nuanced understanding of this phenomenon. We do so by detailing how the meanings that central and local management accountants attach to a CERP system and the attitudes they develop toward it, relate to how the system was developed, who "controlled" it, its ideological underpinnings, and the kinds of expertise that were inscribed in the system.

We demonstrate that technology adopts an ambivalent role for public sector management accountants and conclude that the implications of digital technologies, such as cloud-based systems (see Moll & Yigitbasioglu, 2019), differ not only between organizations, but also affect management accountants differently within the same organizations. Our findings hence emphasize how difficult it is to predict the effects that a specific technology has on the finance function in (and within) different organizations. The present study also reminds us to be cautious about the current

"hype" surrounding digitalization and not to regard digital technologies as "universal" enablers of the finance function (see also Quattrone, 2016). The case of PubliCo illustrates that while some of the features of cloud-based technologies (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019) can indeed be beneficial for central management accountants, those technologies also create challenges for local management accountants. Our case thereby exemplifies the limitations of cloud-based systems that are especially pertinent for local management accountants who are used to developing and controlling *their* local systems to ensure that they cater for the specific needs of their units, while supporting them in performing a local business partner role. We show that CERP systems are underpinned by a specific ideology that can undermine how local management accountants frame the role of technology for their work, with CERP systems seeming to align more with the beliefs and assumptions of central management accountants focusing more on the consolidation of standardized and integrated systems. As such, the case of PubliCo demonstrates that the dynamics and tensions of the technology power loop are not only shaped by different actors' competing for control over systems (see Newman & Westrup, 2005), but also by ideological shifts (see Scarbrough & Corbett, 1992) concerning the role of these systems within the organization or specific functions, such as the finance function in this case.

Accordingly, our study extends prior work that has highlighted differences in the impact of new technologies on central and local management accountants (Goretzki et al., 2018; Heinzelmann, 2017; Newman & Westrup, 2005). We show that off-the-shelf cloud solutions can raise tensions in large and diversified public sector organizations and pose challenges especially for local actors who do not have full control over these systems and are thereby limited in their attempts to mobilize them in the context of their specific expertise. While we acknowledge that problems around customization may also arise for on-premise standardized ERP systems (see, e.g., Wagner et al., 2011), research demonstrates that these problems can be mitigated through collaborative development processes whereby local accountants mobilize expertise associated with their local systems to exert influence on the design of a new central system (Goretzki et al., 2018). This seems more difficult in a cloud-based environment wherein adjustments are expensive, difficult, and require formal approval. Consequently, this not only frustrates local management accountants but also undermines their expertise (Scarbrough & Corbett, 1992), which is often ignored in a cloud environment where it is not always possible to use local systems as knowledge transformation devices (Carlsson-Wall et al., 2019; Goretzki et al., 2018; Kilfyole et al., 2013). However, in diversified public sector organizations especially, flexibility in designing systems is crucial for local management accountants to support decision-making and control processes. Constraining this flexibility may undermine the developments that local public sector management accountants achieved during the NPM movement—inhibiting a desired business partner mentality (Paulsson, 2012). Cloud technology runs the risk that, at the local level, management support reverts to an "old fashioned" approach whereby accounting processes are standardized, cumbersome, and not accommodating of local needs. Coping with an inflexible cloud-based system might therefore add to the list of challenges that public sector management accountants experience when trying to be(come) business partners (Holmgren Caicedo et al., 2018; Mack & Goretzki, 2017; Paulsson, 2012).

We furthermore contribute to the literature by providing in-depth empirical insights into the organizational consequences of cloud-based systems that form a new and hitherto rarely discussed phenomenon in the field of (public sector) accounting (Becker & Heinzelmann, 2017; Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019; Strauss et al., 2015; Yau-Yeung et al., 2020). The notion of commodified systems is taken to a new level by CERP systems and their SaaS delivery method. These technologies involve a greater degree of standardization and provide fewer possibilities for localization. Therefore, to a greater extent, public sector organizations need to adapt to these systems and not vice versa, which contributes to a loss of authority with respect to system design. In this sense, the introduction of a CERP system can reflect what we refer to as an ideological shift; a shift from a focus on local *production* of tailor-made information toward an organization-wide *consumption* of prefabricated information. This shift can be linked to what Quattrone (2016) described in the context of the increasing digitalization of management accounting: "We nowadays consume accounting figures […] as we buy groceries in supermarkets, without knowing how the food has been processed, technically alienated from the manufacturing process, with labels there to reassure us about the ingredients but nothing about whether these are the right nutrients for us" (p. 120). Our findings illustrate that central

management accountants (in contrast to the local management accountants) seem to feel more comfortable with, and even enabled by, consuming prefabricated information produced by externally developed and maintained CERP systems. Cloud-based computing thus has advantages for some parts of an organization or for certain actors but may not be suitable for all organizational units' or actors' needs. Considering public sector management accountants, our study demonstrates that in the wake of the implementation of new digital technologies, a reversion back to a bean counter role may occur for some (local) management accountants—especially those who are used to being involved in system development processes, viewing those systems as important enablers of their role performance. Our study thus illustrates that role changes in the work of management accountant in the context of new (digital) technologies are not as straightforward as is sometimes depicted in the current digitalization discourse (see also Bhimani & Willcocks, 2014; Moll & Yigitbasioglu, 2019).

Future empirical research is needed to better understand the intricacies of implementing and working with cloud-based solutions. For example, what are the implications for power relations and management control when the IT infrastructure is located outside an organization? How should public sector organizations account for vendor risk? How should management control systems be designed to take into consideration a vendor's accountability? Future studies that address the implications, uses, risks, and control features in this new setting, and that further explore the extent to which CERP systems produce different organizational effects compared to traditional ERP implementations, are needed. Furthermore, cloud technology implies that external parties, such as vendors or service providers, gain a larger role in the technology power loop and thus have a greater impact on the role of management accountants. Accordingly, we call for future research on the relationship between external parties and management accountants during system development processes. Finally, we see a need for greater theorization of CERP implementations, especially in terms of notions such as time and emotion. As our case demonstrates, CERP systems affected how management accountants spent their time (Feldman et al., 2020), changed development rhythms (Ancona et al., 2001) and caused feelings of frustration and antipathy (Baxter et al., 2019; Carlsson-Wall et al., 2021). Accordingly, we encourage future research that more deeply probes the temporal and emotional dimensions of CERP developments and implementations.

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NOTES

[1] Source: PubliCo internal documents.
[2] In some cases, two individuals were interviewed together.

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Research data are not shared.

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APPENDIX

TABLE A1 Interview details

Interview	Role	Department	Date	Time
1	System Developer 1 & System Developer 2	Headquarters	05/02-2018	73 min
2	Education Director	Headquarters	14/02-2018	63 min
3	Management Accountant 1	Division 5	02/03-2018	72 min
4	System Developer 1 & System Developer 2	Headquarters	06/03-2018	75 min
5	Finance Director 1	Headquarters	09/03-2018	73 min
6	Finance Director 2 & Management Accountant 2	Headquarters	12/03-2018	70 min
7	Finance Director 3	Headquarters	14/03-2018	69 min
8	Management Accountant 2	Headquarters	14/03-2018	71 min
9	Management Accountant 1	Division 2	15/03-2018	61 min
10	Management Accountant 2	Division 2	15/03-2018	53 min
11	Management Accountant 1 & Management Accountant 2	Division 1	19/03-2018	69 min
12	Management Accountant 3	Headquarters	16/04-2018	56 min
13	Management Accountant 1	Division 4	18/04-2018	65 min
14	Management Accountant 2	Division 4	18/04/2018	47 min
15	Management Accountant 3	Division 1	19/04/2018	58 min
16	Management Accountant 1	Division 3	24/04/2018	70 min
17	Management Accountant 3	Division 2	24/04/2018	64 min
18	Management Accountant 2	Division 5	25/04/2018	65 min
19	Management Accountant 4	Headquarters	26/04/2018	40 min
20	Management Accountant 2	Division 3	08/05/2018	51 min